

Into the Womb -I want to be born again-

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Figure 1: Into the Womb: Outside system including ball chair and inside overview of the system of fetus experience

Abstract

“Into the Womb” is an XR content with a simulated experience of being born again. The author who has a developmental disability expresses her hope that if she could be reborn without her disorder, she would feel universal love. The work raises social issues about women with developmental disabilities, who have lost love, which is the most important human emotion, due to trauma.

The protagonist, who suffers from a developmental disability at the beginning, is verbally tricked by people and expresses her difficulty in living, having been traumatized by love and other things. The story begins with a scene of the world from the perspective of a girl with developmental disabilities in a night city called Kabukicho in Shinjuku, Tokyo. Participants will see the transition from the dark night city back into the womb of their mothers, experiencing a simulated escape from a society filled with contradictions. They can hear the mother’s words and the encouraging words around them, which confirm their love. Then, they realize that they are being born again by moving their hands inside the womb to be born again. In the end, they assimilate with the fetus and are enveloped

in light, and they are born again and ‘alive’ in the world. When they finish the experience and remove the headset, they feel a world full of hope after they have been born again.

This work aims to help people feel free from trauma and have heart-warming feelings by going through a simulated experience of being born again.

CCS Concepts

• Applied computing → Arts and humanities; Media arts.

Keywords

Autism and developmental disabilities, Virtual reality, Fetus, Social understanding, Womb

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1 Introduction

This work is a virtual reality (VR) content of being born again (Figure 1), inspired by the experiences of people who do not fit into society because of their developmental disabilities. In Japan, girls with developmental disabilities gather in the nightlife of downtown areas and repeatedly engage in delinquency and prostitution, which has become a social problem. This work is a fetal return VR

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which proposes that if the girls were born without developmental disabilities, their lives could be brightly illuminated.

The following previously developed VR content is themed around the fetus. Mummy Tummy [Kosaka et al. 2011] is a pregnancy experience that aims to convey the difficulties and joys of pregnancy. It uses air pressure and other means to represent the movement of the womb. The approach of amplifying experiences closely related to the body through haptic sensory stimulation is considered useful. Perinatal Dreaming [Kuchelmeister et al. 2023] is a work that focuses on creating an immersive visual experience around the theme of the womb and birth. This is just audio-visual content and not a multisensory experience that includes space and haptic sensations, such as being enveloped in the womb. The authors previously proposed a fetal experience of returning to the womb by wrapping the people in a contractile cloth and viewing audio-visual VR content about the inside of the womb [Kimura et al. 2020], [Kimura et al. 2021]. However, people are only wrapped in the cloth, and it is not a haptic experience synchronized with the VR video.

This research proposes a multi-sensory experience of being in the womb using air cushions that can be dynamically controlled in accordance with VR videos as a means of alleviating social anxiety. It provides an experience of being born again for people with developmental disabilities and others. While much conventional VR content on the fetus has been aimed at preparing for pregnancy and childbirth or conveying their greatness, the novelty of this research is that it utilizes the experience of returning to the fetus to simulate a new start for people who are anxious about their own lives.

2 Method

2.1 Overview of the Experience

This VR content is an experience of being born again. It consists of VR videos of returning to the fetus and moving synchronously with the fetus, mystical sounds, and haptic sensations like being enveloped in the womb. The content lasts approximately 2 minutes.

2.2 VR Audio-Visual Contents

In the VR video, the experience begins from the perspective of a woman with a developmental disability. People who experience this content feel the trauma and contradictions of this world held by the girls who have no place at home or school and gather in the nightlife of Tokyo, Japan. The night city and the womb gradually synchronize, and the experience in the womb begins. In the scenes in the womb, people first see the whole body of the fetus. The fetus moves closer to the people, and they are assimilated into the fetus. The arm movements of them are synchronized with those of the fetus. The movement of the fetus's arms is restricted to represent the narrow environment inside the womb, and ripples of amniotic fluid occur by using the shader effects when tapping the inner wall of the uterus with the arms. At the end of the experience, the fetus separates from the people and is encased in light to represent being free from past trauma and born again.

The sound of this work is composed and arranged on our own to be relaxing music using piano and heartbeat sounds to match the theme of returning to the womb and starting life over. The soprano singer's voice is recorded and incorporated into the music to make it sound as if the mother is speaking to the fetus.

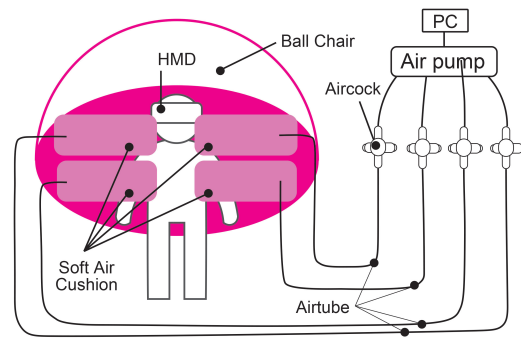


Figure 2: Overview of the haptics system.



Figure 3: Haptic sensation using air cushions.

2.3 Haptics System

We develop a haptics system that aims to give the sensation of being surrounded by affection in the mother's womb. The overview of the system is shown in Figure 2. While experiencing the audio-visual content, people sit on a spherical chair and feel the sensation of being compressed by the inflated air cushion set in the chair, and feel as if they are encased in a small space like the womb (Figure 3). By controlling the air pump in conjunction with the VR contents, the air cushion inflations change according to the experience phase.

3 Conclusion

In this paper, we describe the work which is a virtual experience of being born again. The audio-visual VR content and dynamic haptic sensations are used to represent the sense of returning to the womb. We hope that this work will give people the feeling of starting a new life and living brightly and strongly.

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